Predicting US Skin Cancer Status

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Abstract

This project aims to predict skin cancer diagnosis status (benign or malignant) using statistical learning models applied to a dataset of 50,000 observations and 50 predictors. The analysis includes data cleaning, imputation, model construction, and performance evaluation. Missing values were handled using the MissForest imputation method to preserve relationships among variables.

We compared logistic regression, Linear Discriminant Analysis (LDA), and Random Forest models based on test accuracy and interpretability. While Random Forest models exhibited overfitting and poor generalization, linear models performed more reliably. The final model, LDA, achieved the highest test accuracy of 0.6059. Key predictive features include skin tone, family history, sunscreen usage frequency, immunosuppression status, and tanning bed use. Although predictive accuracy is modest, the model provides useful insights for early skin cancer risk assessment.

1. Introduction

Skin cancer is one of the most common types of cancer in the United States and also one of the most preventable types of cancer. Despite this, it remains a serious health concern when malignant cases are not identified early. Although a majority of skin lesions are benign, malignant skin cancers such as melanoma can develop quickly and lead to severe consequences if they are not left untreated. Therefore, early detection and getting an accurate assessment of the risk are crucial in improving patient outcomes. As more medical and epidemiological data become increasingly available worldwide, statistical methods have emerged as useful tools for analyzing these data and recognizing patterns related to skin cancer risk.

This project is based on a Kaggle competition that focuses on predicting skin cancer status (Benign vs. Malignant) that uses a large dataset. The dataset is designed to reflect real-world dermatological and epidemiological risk factors. The training data includes 50,000 observations with 50 predictors and a binary response variable that tells us the cancer status, while the test dataset contains 20,000 observations without response labels. These predictors cover a broad range of factors. These predictors cover a broad range of factors, including demographic characteristics, environmental and UV exposure, sun protection behaviors, biological traits, and even lifestyle choices. Together, they try to capture the many factors that can influence an individual's risk of developing skin cancer.

The goal of the competition was to build a model that could attempt to distinguish between benign and malignant cases using the training data. We would then generate predictions for the test dataset for evaluation. By framing skin cancer risk as a binary classification problem, the competition clearly demonstrates how statistical techniques can be applied to a meaningful real-world health problem. It also shows the potential for data-driven methods for early detection efforts and improving decision-making in medical and public health settings.

2. Data Analysis

2.1 Dataset Overview

The dataset consists of 50000 observations and 50 predictors. After loading the dataset, we began with the removal of the index column to eliminate any non-predictive noise. Subsequently, we ensured that all the predictors were converted to the correct variable type.

A. Numerical Predictors

The dataset contained a multitude of numerical variables, ranging from demographic variables, environmental variables, and physical measurements. Figure 1 shows the distributions of the variables. Most of the variables seem to be fairly equal between categories, but age was a standout as we see a large discrepancy between the two categories.

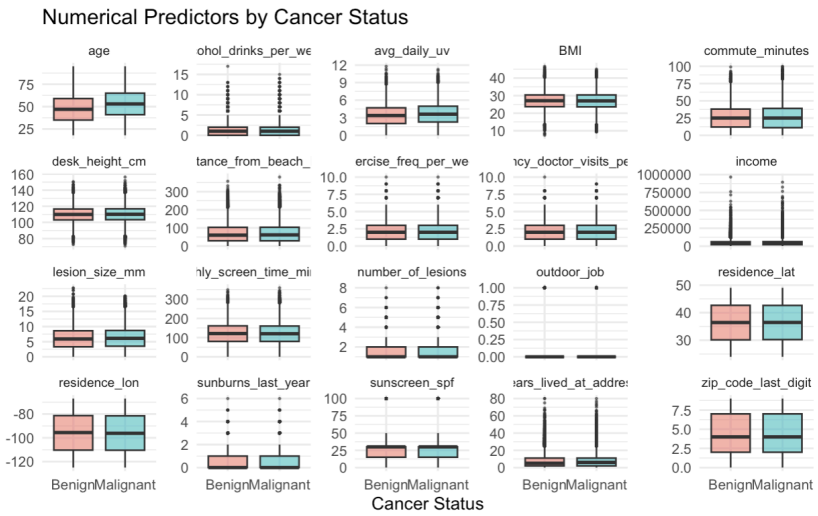

Figure 1: Numerical Predictors by Cancer Status

B. Categorical Predictors

Figures 2 and 3 show the two categorical variables, skin_tone and gender. These variables were examined at every level within each category. The frequency distribution of the factors is essential and can lead to instability in modelling predictions. Gender was pretty equally distributed, while skin_tone had some imbalances.

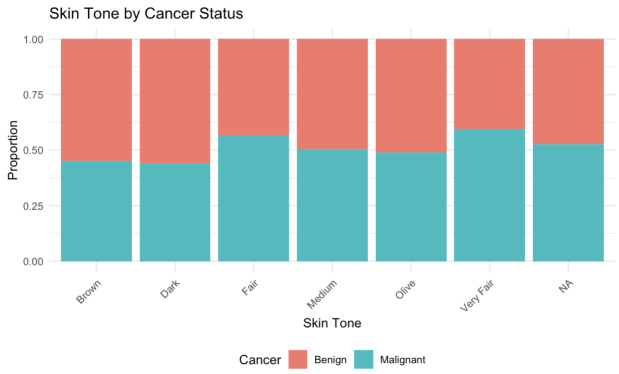

Figure 2: Skin Tone by Cancer Status

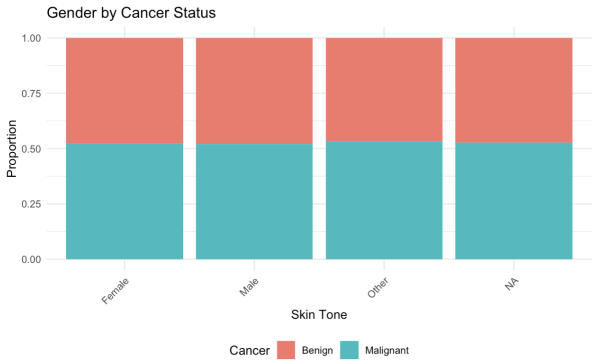

Figure 3: Gender by Cancer Status

2.2 Missing Value Handling

The dataset consists of missing NAs distributed across both numerical and categorical predictors. Prior to model fitting, we examined the extent of missingness for each variable to ensure that no single predictor exhibited an excessively high proportion of missing values that would warrant exclusion.

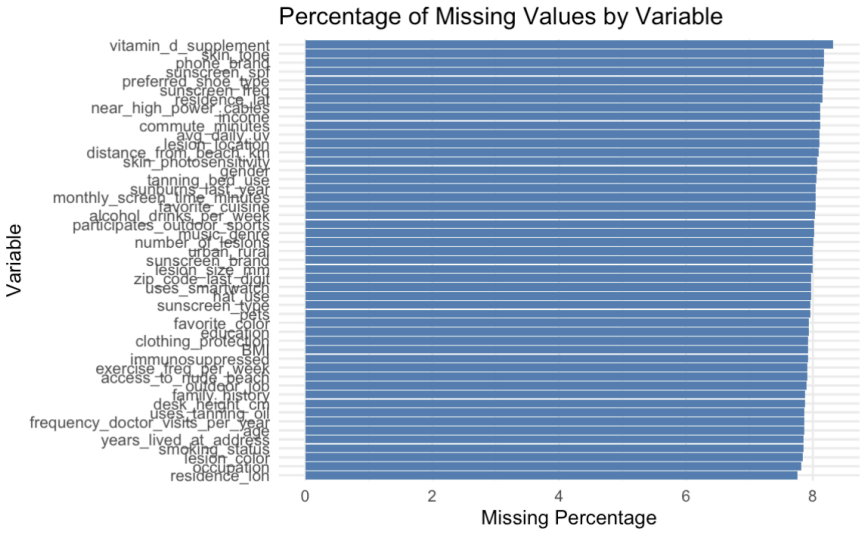

Figure 4: Percentage of Missing Values by Predictors

As shown in Figure 4, the percentage of missing values across predictors was relatively consistent, with approximately 7–8% missingness overall. To address the missing values

visualized without reducing the sample size, the missForest algorithm was utilized for imputation. This non-parametric approach estimates missing values using Random Forests, allowing the model to account for complex, non-linear dependencies between variables. This method was selected over simple mean or median imputation to minimize bias and preserve the underlying variance structure of the dataset.

2.3 Class Balance and Target Analysis

An analysis of the response variable was conducted to establish a baseline understanding of the disease. The class distribution was found to be 52.26% for positive cancer cases (Malignant) and 47.74% for non-cancer cases (Benign). Since these were roughly equal, it indicates a balanced dataset, allowing the usage of standard model evaluation.

2.4 Variable Selection

Given the large number of predictors in the dataset (50 variables), we decided to perform a variable selection prior to model construction to reduce dimensionality. Using all variables increases the risk of overfitting when using models like trees.

To address this, we used variable importance (VIP) analysis as an exploratory tool to identify the most influential predictors. Variable importance measures evaluate how much each variable contributes to predictive performance by assessing the change in classification accuracy when the variable's information is disrupted. Variables that lead to larger decreases in accuracy are considered more informative.

All predictors were first evaluated and ranked according to their Mean Decrease in Accuracy scores. This ranking allowed us to compare variables on a common scale and identify

those with the strongest association with cancer status, without relying on model-specific assumptions such as linearity or distributional form.

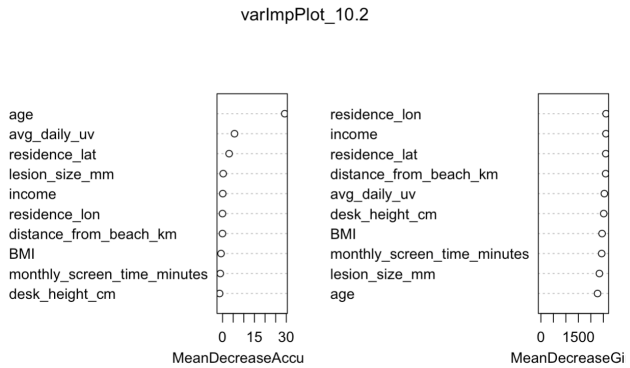

Figure 5: Top 10 Predictors

Based on this ranking showed in figure 5, we selected our top 10 variables: age, avg_daily_uv, residence_lat, income, distance_from_beach_km, residence_lon, desk_height_cm, BMI, monthly_screen_time_minutes, and lesion_size_mm.

2.5 Linear Feature Associations

To verify these results, we performed a secondary assessment using Linear Discriminant Analysis coefficients. Unlike the Variable Importance analysis in Section 2.3, which captures non-linear relationships and complex interactions, LDA creates weights based on a feature's ability to separate the data linearly.

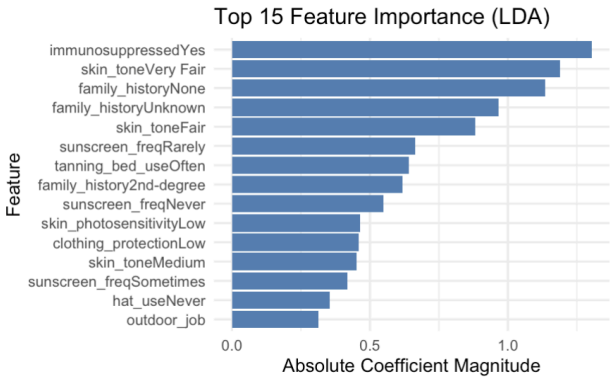

Figure 6: Linear Feature Associations

As illustrated in Figure 6, LDA identified notably different features than those selected by the VIP. While the Random Forest prioritized variables like age and avg_daily_uv, LDA assigned higher absolute weights to variables such as immunosuppressed and skin_tone.

This divergence between the two methods of ranking importance provides a crucial insight: the risk factors for skin cancer do not follow a simple linear trend. The variables identified by VIP likely influence cancer risk through a plethora of different interactions, whereas the LDA top features represent the few factors that have a direct, monotonic relationship with the disease. This complexity supports our decision to explore non-linear modeling techniques in the subsequent analysis.

3. Methods and Models

Prior to developing our models, the dataset was divided into training and testing sets. All models were fit on the training data and evaluated based on their ability to correctly classify cancer status prediction on the testing set. Using this framework, we compared the predictive performance, defined by model accuracy, of several classification methods, including logistic regression, LDA, and various random forest models.

3.1 Logistic Regression

Logistic regression models are simple yet effective, performing well with many predictors, while producing easily interpretable results. Given that the dataset has 50 predictors, this logistic regression model is an efficient choice, and was implemented using the glm() function with the binomial response being malignant or benign. The model was trained on the cleaned dataset.

Variable selection was evaluated on accuracy. When models were tested using a reduced number of predictors, the accuracy consistently decreased, so all predictors were retained in the final model. Logistic regression assumes a linear relationship between predictors and the log-odds of cancer, making it an appropriate and interpretable baseline before attempting more flexible modeling approaches. The final logistic regression model with all predictors achieved an accuracy of 60.575%.

3.2 Linear Discriminant Analysis

LDA is another efficient classification method that assumes normality and equal covariance between classes, performing best when class separation is linear. When the boundaries of separation are more complex, the LDA model will perform worse. Similar to the logistic regression model, the LDA model was trained on the cleaned training dataset, and performed best with all predictors included. The final LDA model with all predictors achieved an accuracy of 60.59%.

3.3 Random Forest

Random Forest is an ensemble classification method that builds a large number of decision trees using bootstrap samples of the data and aggregates their predictions through

majority voting. By randomly selecting a subset of predictors at each split, Random Forest reduces correlation among trees and can model complex nonlinear relationships while remaining robust to noise and multicollinearity.

Model with All Predictors:

We first trained a Random Forest model using all 50 predictors from the cleaned training dataset. To improve model stability, we set the number of trees to ntree = 800, as increasing the number of trees generally reduces variance and stabilizes predictions without substantially increasing computational cost.

For the number of predictors considered at each split, we used mtry = floor((p − 1) / 3), where p is the total number of predictors. This choice is a well-supported rule of thumb for classification problems with many variables, balancing model flexibility and stability. Exhaustively tuning all possible mtry values would require training dozens of large Random Forest models and was computationally unnecessary for this dataset.

Despite these choices, the Random Forest model exhibited clear signs of overfitting. The model achieved perfect accuracy on the training set but performed substantially worse on the test set, with an accuracy of approximately 59.5%. This large gap indicates that the model memorized patterns in the training data but failed to generalize to new observations.

Model with Top 10 Predictors:

To reduce model complexity and address overfitting, we next restricted the Random Forest model to the top 10 predictors selected using our variable selection based on the variable importance measures, which ranked predictors based on their Mean Decrease in Accuracy to

measure how much classification performance declines when a variable is permuted. Variables with larger decreases were considered more influential and retained for the reduced model.

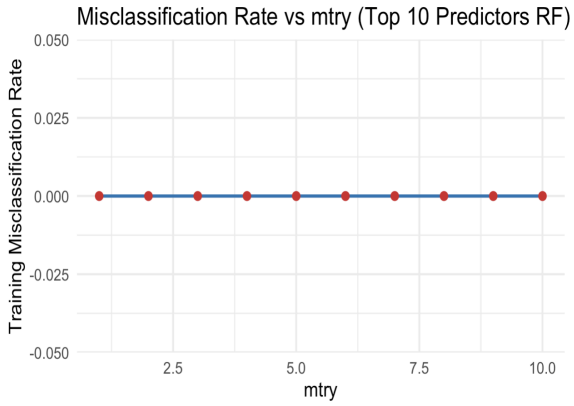

Figure 7: Misclassification Rate based on mtry 1:10

Using this subset of predictors, we retrained the Random Forest and tuned the mtry parameter by evaluating values from 1 to 10. However, all configurations achieved 0% training misclassification, indicating that the model continued to perfectly fit the training data regardless of the mtry choice. As a result, tuning mtry did not meaningfully improve generalization. The reduced Random Forest model achieved a test accuracy of approximately 55.4%, which was lower than the full-predictor Random Forest and did not outperform simpler models.

Result Analysis

To compare model performance, we evaluated testing accuracy across all fitted models. The results are summarized as follows:

- GLM achieved a testing accuracy of 0.60575
- LDA achieved a testing accuracy of 0.60590
- Random Forest (all predictors) achieved a testing accuracy of 0.59545
- Random Forest (top 10 predictors) achieved a testing accuracy of 0.55420

Among all models, LDA produced the highest testing accuracy, although the difference between GLM and LDA was small. Both linear models outperformed the Random Forest models, which showed signs of overfitting and weaker generalization performance on the test set. Based on these results, LDA was selected as the final model.

4. Discussion and Limitations

In this project, we compared multiple classification models to predict skin cancer status using a high-dimensional dataset with mixed predictors. While nonlinear models such as Random Forest performed well on the training data, they exhibited substantial overfitting and poor generalization. In contrast, linear models, particularly LDA, were more stable and interpretable and achieved the highest test accuracy among the models considered.

The final LDA model identified several important risk factors, including skin tone, family history, sunscreen usage frequency, immunosuppression status, and tanning bed use. These findings are consistent with established clinical and behavioral risk factors and indicate that simple linear models can capture meaningful patterns in large observational datasets.

However, the analysis has notable limitations. The presence of class imbalance in the dataset likely contributed to the modest predictive accuracy observed across all models and reduced performance in identifying malignant cases, which are clinically more important to detect. In addition, LDA relies on the assumption that class covariance matrices are equal, and violations of this assumption may weaken its discriminative ability. The use of MissForest imputation, while effective for handling missing values and preserving complex relationships, may also have altered underlying data patterns and introduced additional uncertainty, potentially affecting model stability and reliability.

5. Conclusion and Recommendations

In this project, we compared several classification models to predict skin cancer status. We used a dataset comprising various risk factors and found that simpler models performed more reliably than the more complex ones. Although random forest models yielded strong results on the training data, they tended to overfit and generalize poorly. On the other hand, LDA and logistic regression gave us more stable results on the test data. LDA ultimately received the highest testing accuracy.

One benefit of using LDA is that it allows us to understand which predictors most affected the model's decisions. Some variables like skin tone, family history of skin cancer, sunscreen usage, and immune-related conditions came out to be the most influential ones. These factors actually align with what is already known about skin cancer risk. This tells us that the model was actually showing us meaningful patterns rather than random noise. Even though the overall accuracy was not very high, the model still gave us some useful insight into how different risk factors contribute to classification.

Looking back, there are several limitations to be considered. The dataset was synthetically generated. This means it may not fully represent real clinical data. Also, missing values were filled in, which could have affected the structure of the data. LDA also relies on certain assumptions that may not hold perfectly. Even with these limitations, the final model gave us a pretty reasonable balance between what we can infer and performance, all while showing us the challenges of applying statistical learning methods to real-world health problems.

References

Almohalwas, Akram. Lecture Slides. STATS 101C, University of California, Los Angeles.